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Instruction Date	:	**17 October 2017**
Our Ref Number	:	**GSS 325376**
London Ref	:	**SS**
Report Date	:	**1 August 2018**

SURVEY REPORT

THIS IS TO CERTIFY

that at the request of Underwriters concerned, survey has been held on the steel built, twin screw, motor driven, offshore support vessel: -

"BOURBON EVOLUTION 802"

GT 6146 of Luxemburg

while at Carena Ship Repair, Abidjan, for the purpose of ascertaining the cause, nature and extent of damage alleged to have been sustained in the following circumstances:

10 JANUARY 2017: CRANKSHAFT DIESEL GENERATOR # 3 BREAKDOWN

It was reported during normal operation, two engines DG3 and 5 were working in parallel on load due to cranes operation. DG3 lubricating oil filters differential pressure high alarm occurred, and the engine was stopped by emergency stop.

Oil filters were inspected and metal particles were found inside the filters.

Crankshafts were inspected and found in good condition. Crankcase was inspected. All connecting rods were inspected. Unit 4RB connecting rod close to connecting rod cap was found discoloured. Unit 4RB and 4LB connecting rod caps were removed. Connecting rod bearings were found damaged and crankpin No.4 found seized

All our assignments are subject to our Terms and Conditions that are available on request or here;
http://braemartechnical.com/files/23-Standard-Terms-Conditions-2014.pdf

LLOYD'S REGISTER · LRQA
ISO 9001 · ISO 14001
OHSAS 18001

For further particulars please refer to log books and/or statements, the foregoing being from verbal statements made to us by a representative of the Owners and review of documentation presented by them. The logbooks were sighted on this occasion.

19 JANUARY 2018

On the above occasion and subsequent occasions, the undersigned surveyor attended onboard and in the company of the following found, noted and was advised as follows:

ATTENDING THE SURVEY

Name & Position	Representing
Capt. Bonaci Tonci	Owners
Capt of Bourbon Evolution 802	
Mr. Drelich Andrzej	Owners
Chief Engineer of Bourbon Evolution 802	
Name & Position	Representing

VESSEL’S DESCRIPTION

The vessel is an offshore supply vessel. It was built during 2012 in Zhejiang Shipbuilding Co. Ltd, China and has a loaded service speed of 12 knots.

Principal Particulars

Owners	-	Bourbon Evolution 802 SNC
Manager	-	Bourbon Offshore Surf
IMO Number	-	9629835
Class	-	Bureau Veritas
Flag State /Registry	-	Luxemburg

Principal Dimensions

Length (OA)	-	100.20 m
Length (BP)	-	95.60 m
Moulded Breadth	-	21 m
Moulded Depth	-	8 m
Max Draft	-	6.35 m
Gross Tonnage	-	6146
Net Tonnage	-	1843
Deadweight	-	4849 tonnes

Propulsion Prime Mover

Make	-	Cummings
No.	-	3
Type	-	Diesel
Model	-	MCR 5052
Cylinders	-	V16
MCR	-	5052 kW @ 1800 RPM
Engine Builder	-	Cummings
Fuel Type	-	MGO

Propulsion Unit

Propeller	-	3 Solid Azimuth Thrusters
Drive	-	Via reduction gearbox

Class & Statutory Certification

- The vessel remains classed with Bureau Veritas.
- Document of Compliance issued by Bureau Veritas on 22 March 2017 valid until 23 March 2022
- Safety Management Certificate issued by Bureau Veritas on 12 December 2018 valid until 12 May 2018
- The casualty was reported within the company Safety Management System.
- All statutory certificates were noted to be valid
- There were no existing conditions of Class at the time of the incident.

DAMAGE FOUND

On thorough inspection, the following was found

AUX ENGINE # 3	
Maker	Cummings
Model	KTA 50D (MC)
Type	4 stroke, V16
Power Rating	1,235 kW
Serial No.	33177714
Date of Manufacture	2/2009
Run Hrs Since New/overhaul	28645 / 11817

During our attendance on 19 January 2018, the following was noted:

1. The crankcase doors for no.4 units (Right & left banks) were opened for inspection and bearing caps and shells removed.

2. The bearing caps and bearing shells on both bank were darkened by heat damage. The bearing shells were deformed and surfaces wiped.
3. The connecting rods were also noted to have heat damage.
4. The crankpins were found to be severely darkened from overheating and severely scratched.
5. Oil sample taken from the crankcase showed a large presence of metallic particles.

We reattended the vessel on 24 February 2018 and the following was noted:

1. The owners subsequently ordered and installed a new diesel generator engine to replace the damaged DG#3. At the time of our second attendance, the installation was completed, and the engine was being commissioned.
2. The damaged old engine had been taken out of the vessel to the Carena Ship Repair yard engine workshop where it was sighted.
3. The engine was partially dismantled to enable inspection of the damage in the crankcase to be undertaken.
4. The nos 4 & 5 bearings shells were noted to be severely scored and scratched with the copper underlay exposed in the case of no. 5 unit.
5. Inspection of the corresponding position on the crankshaft found the crankpin severely grooved and scored over approximately 80% of the surface area at unit 4.
6. The crankshaft was damaged beyond economic repair and would have to be replaced.

ALLEGATION

No formal written allegation of cause has been made in respect of this damage. It is understood that if a claim is to be made, the Owners will notify their Brokers of the allegation of cause. In view of the foregoing, the damage now found and reported upon are noted Without Prejudice to Underwriters’ liability and the Owners Representative has been so advised.

<u>Remarks</u>

Even though there is no formal allegation, the following has been noted:

1) The Owners reported in their email dated 7 March 2018 from Douchras Jean-Patrick that: **Quote** *Makers has reported that engine block may have a defect (from factory or yard) which generated this breakdown. Makers advised to not repair it because the same issues may happen again* **end of Quote***.*
2) It is noted, as also indicated in the same email and our Advice No.1, that this is the second time the failure has occurred around the same 10,000 running hours (maximum recommended hours being 24,000 hrs before overhaul) within 2 years.
3) In view of the above, we have requested that the manufacturers provide further information to confirm their position. If confirmed, we could possibly then conclude on what is the most likely cause of the damage.

PERMANENT REPAIRS CARRIED OUT

The damaged DG#3 was replaced with a reconditioned one which was satisfactorily commissioned and reported to be operating satisfactorily.

STATUS OF REPAIRS

Repairs are completed.

GENERAL SERVICES & ACCESS

1. Vessel docked stern to, for these repairs with tug assistance, pilot’s advice, line and gangway services.
2. Provided the following:

- Opening up of manholes in way of damage and closing with new gaskets on completion of repairs
- Cranage
- Disposal of dirt and debris of repairs
- Garbage services
- Security

3. All new and disturbed material coated as original.
4. All new and disturbed material tested and proved satisfactory to satisfaction of all parties including Class.
5. All new and disturbed material tested and proved satisfactory to satisfaction of all parties including Class.

ESTIMATED COST

Since no further work was undertaken on the damaged DG#3 our estimated cost of repairs (taking into consideration the work done in removal and partial disassembly of the damaged engine) was amended from our Advice no.2 to USD220,000, subject to variation when final invoices are received.

ACCOUNTS

The accounts are approved by us, without prejudice, subject to Underwriters liability and adjustment in the usual manner being considered fair and reasonable as indicated below. A query was raised on one account item which was presented without an invoice. When a response is received on the said query it will be reviewed in an addendum to this report.

1. BDO

Invoice no. 2018200018	*22 March 2018*	*USD*	*137,871.34*

The above is for new engine by BST 33173664 for the Bourbon Evolution 802.

Sum approved as fair, reasonable and casualty related USD 137,871.34

2. BOS

Invoice no. none	*18 February 2018*	*USD*	*5,500.00*

The above account is for supervision by Stephaine LEROUX (technician) during repairs of the casualty

Sum approved as fair, reasonable and casualty related USD 5,500.00

3. CARENA

Invoice no. **C**092/18	*26 March 2018*	*EUR*	*52,692.00*

The above works carried out between 02/01 and 31/01/2018 @ Carena Shipyard, Abidjan , Cote D’Ivoireourbon Evolution 802.

Sum approved as fair, reasonable and casualty related EUR 52,692.00

4. CARENA

Invoice no. **C**092/18	*26 March 2018*	*EUR*	*52,692.00*

The above for works carried out between 02/01 and 31/01/2018 @ Carena Shipyard, Abidjan, Cote D’Ivoire.

Sum approved as fair, reasonable and casualty related EUR 52,692.00

5. MECATLANTIC

Invoice no. **1970045**	*14 March 2018*	*EUR*	*1,500.00*

The above for issue of EIAPP for a new engine.

Sum approved as fair, reasonable and casualty related EUR 1,500.00

6. MECATLANTIC

Invoice no. **1970044**	*14 March 2018*	*EUR*	*17,816.09*

The above for supervision of works by technician Ivo Gugge in Cote D’Ivoire for replacement of engine on the Bourbon Evolution 802 from 12/02 to 17/02/2018.

Sum approved as fair, reasonable and casualty related EUR 17,816.00

REPAIR TIMES

The vessel was at the repairers solely on account of the damage noted herein, no Owner’s work being carried out concurrent with damage repairs, which took the full period afloat to effect.

SURVEYORS NOTES

1. We were assisted by the technician from Mecatlantic Mr Philippe Raymond.
2. Records indicate that the DG3 had a similar breakdown in February 2015 with the same no. 4 unit being damaged at the time.
3. Due to the restrictions in the engine room, it was agreed that in order for a full assessment of the damage to be done, including examination of the main bearings, especially in way of no. 4 crankshaft unit, it would be better for the DG3 to be taken out of the vessel to shore workshop for disassembly.
4. Alignment of crankshaft and main bearings to also check and an examination of material to be undertaken.
5. A review of the engine room maintenance records indicated that the maintenance schedules were being adhered to. The DG 3 running hours since last overhaul was less than half the recommended 24,000 hrs. The last lubricating oil change was 367 hrs since last overhaul and crankshaft examined on 4.5.17.
6. The technician from Mecatlantic has indicated that it would be faster to provide a replacement engine while the damaged one is carefully examined and repaired if considered a viable option.
7. The manufacturers subsequently indicated (as per email of 7.3.2018 from owners) that the engine had a manufacturing defect and advised that if it is repaired the defect could occur again, the owners as a result, opted not to repair the damaged engine.
8. The manufacturers subsequently indicated that the engine had a manufacturing defect, the owners as a result opted not to repair the damaged engine.

PRINCIPAL DATES

Date	Comment
10.1.2018	Date of incident.
16.1.2018	Examination of damage started by Mecatlantic
19.1.2018	Assessment completed.
12.2.2018	Old engine removed from vessel
14.2.2018	New engine arrived in Shipyard.
15.2.2018	Installation started
22.2.2018	Installation completed.
24.2.2018	Engine commissioned.

DOCUMENTS RETAINED ON FILE

Copies of the following documents are retained by us on file:

1. Vessel general particulars
2. Crew list
3. Safety Management Certificate
4. Document of compliance
5. Incident report.
6. Certificate of registry.

DOCUMENTS REQUESTED

Copies of the following documents have been requested from the Owners:

1. We have requested that engine makers provide confirmation that the block has a manufacturing defect from the factory or yard as indicated in their report.

ATTENDING SURVEYOR:

Nasir Mohamed
Braemar Technical Services Ltd

APPENDED

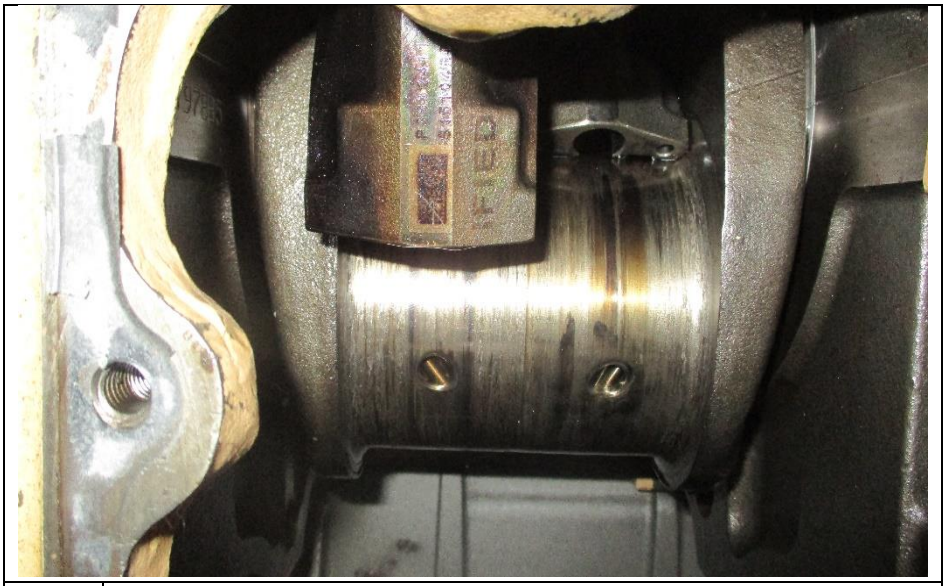
- Selected Photographs

CONFIDENTIALITY: This report is confidential. It is written for the benefit of the addressee but solely for the purpose of assisting in the resolution of the issues with which it deals. It is to be used for that limited and confidential purpose only. In the event of unauthorised release, it and all records or copies of it are to be returned immediately. No reproduction nor repetition nor disclosure of any conclusion in it, nor of any information drawn from it, in whatever form, for any other purpose than that, is permitted without the specific written permission of Braemar Technical Services Ltd.

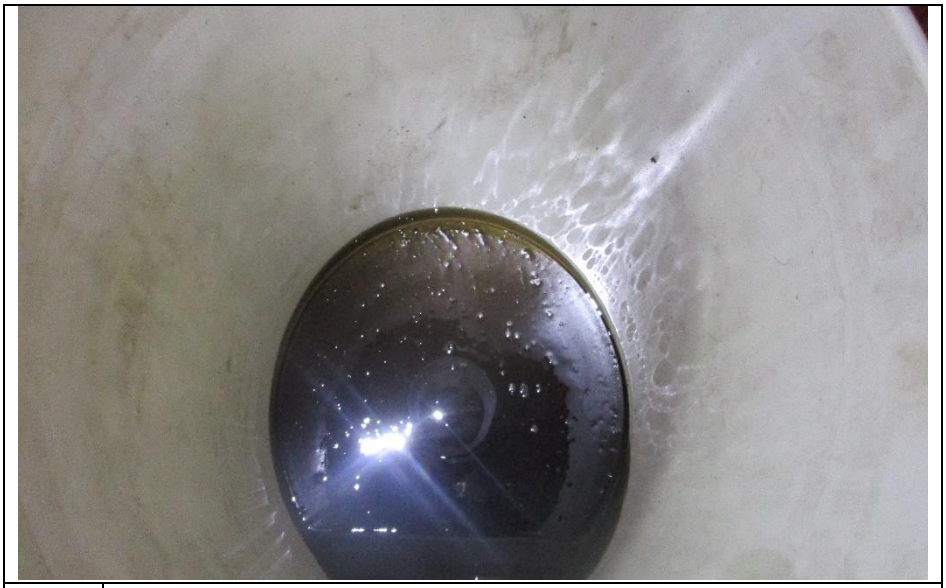
SELECTED PHOTOGRAPHS

1.	No.4 unit RB & LB bearing shells and caps darkened by heat damage. Bearing shells scratched and wied.

2.	Metallic particles on top of lubricating oil filter.



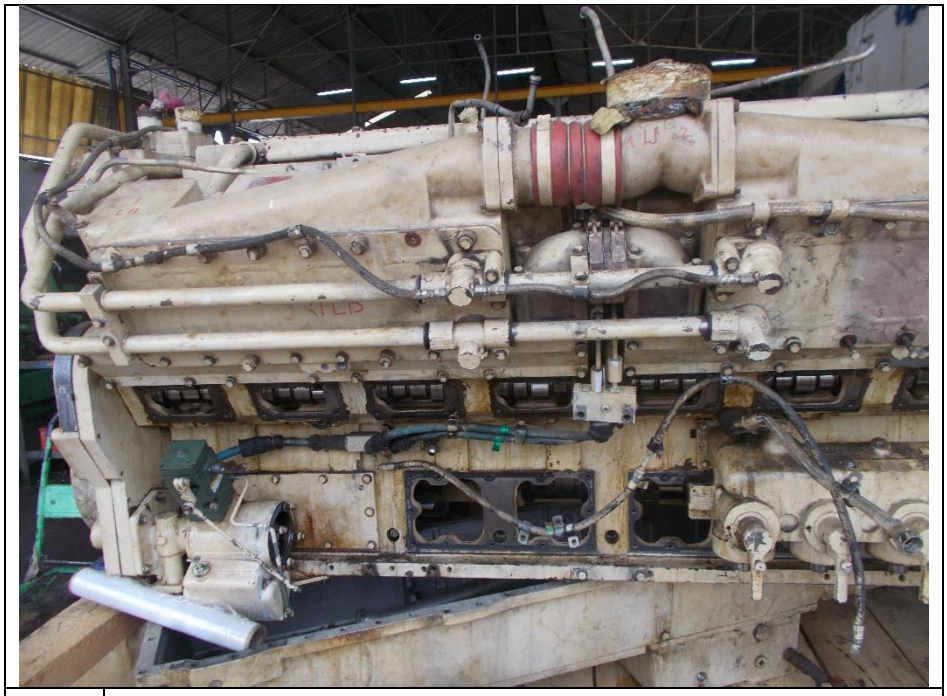
3.	Crankpin journal severely scratched and patches darkening due to overheating.



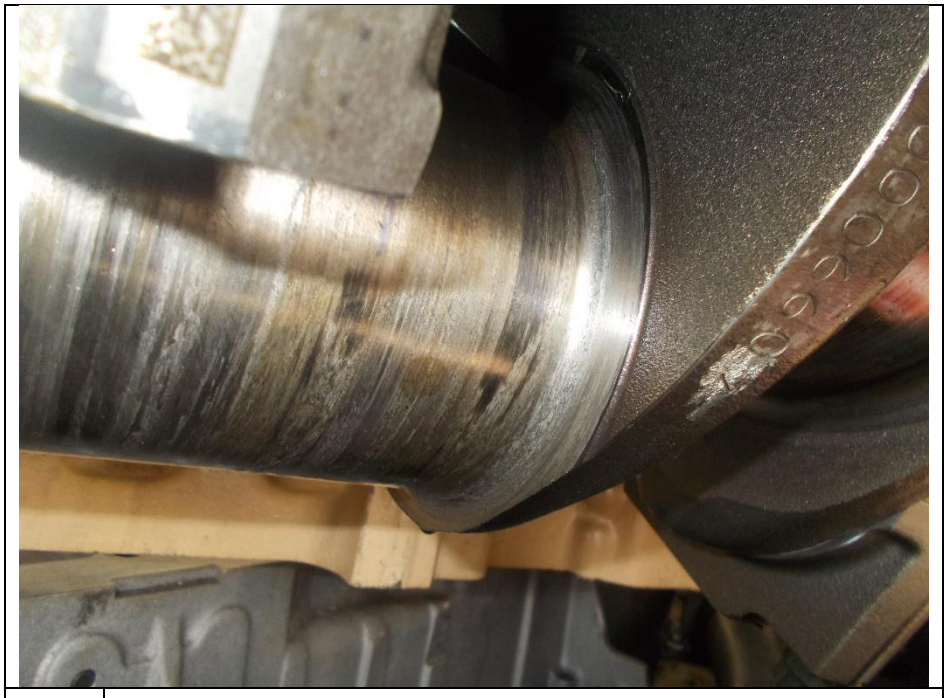
4.	Metallic particles in sample lubricating oil taken from crankcase of DG no.3 engine.

5.	New engine -replaced DG#3-installed on vessel.

6.	New engine in place.



7.	Old DG/E as seen in Carena workshop, partially disassembled for inspection.



8.	Crankpin in way of no.4 unit severely scored and with deep grooves- close up photo taken in Carena workshop.